Rajalakshmi Engineering College

FUNDAMENTALS OF DATA SCIENCE LAB

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EXPERIMENT 15

Understand data preprocessing in Data Science

Experiment 15: Understand data preprocessing in Data Science

Aim

To demonstrate data preprocessing in data science.

Source Code

```
import numpy as np
import pandas as pd
df=pd.read_csv("/content/pre-process_datasample.csv")
df
df.info()
df.Country.mode()
Country
df.Country.mode()[0]
type(df.Country.mode())
pandas.core.series.Series
def __init__(data=None, index=None, dtype: Dtype | None=None, name=None, copy: bool | None=None,
fastpath: bool=False) -> None
df.Country.fillna(df.Country.mode()[0],inplace=True)
df.Age.fillna(df.Age.median(),inplace=True)
df.Salary.fillna(round(df.Salary.mean()),inplace=True)
df
pd.get_dummies(df.Country)
updated_dataset=pd.concat([pd.get_dummies(df.Country),df.iloc[:,[1,2,3]]],axis=1)
updated_dataset
df.info()
updated_dataset.Purchased.replace(['No','Yes'],[0,1],inplace=True)
updated_dataset
```

Output - Experiment 15

Country Age Salary Purchased
0 France 44.0 72000.0 No
1 Spain 27.0 48000.0 Yes
2 Germany 30.0 54000.0 No
3 Spain 38.0 61000.0 No
4 Germany 40.0 NaN Yes
5 France 35.0 58000.0 Yes
6 Spain NaN 52000.0 No
7 France 48.0 79000.0 Yes
8 NaN 50.0 83000.0 No
9 France 37.0 67000.0 Yes
Double-click (or enter) to edit
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
0 France
index is not None, the resulting Series is reindexed with the index values.
dtype : str, numpy.dtype, or ExtensionDtype, optional
Data type for the output Series. If not specified, this will be
inferred from `data`.
See the :ref:`user guide <basics.dtypes>` for more usages.
name : Hashable, default None
The name to give to the Series
Country Age Salary Purchased
0 France 44.0 72000.0 No
1 Spain 27.0 48000.0 Yes
2 Germany 30.0 54000.0 No
3 Spain 38.0 61000.0 No
4 Germany 40.0 63778.0 Yes
5 France 35.0 58000.0 Yes
6 Spain 38.0 52000.0 No
7 France 48.0 79000.0 Yes
8 France 50.0 83000.0 No
9 France 370 670000 Yes
France Germany Spain
0 True False False
1 False False True
2 False True False
3 False False True
4 False True False
5 True False False
6 False False True
7 True False False
8 True False False
9 True False False
France Germany Spain Age Salary Purchased
0 True False False 44.0 72000.0 No
1 False False True 27.0 48000.0 Yes
2 False True False 30.0 54000.0 No
3 False False True 38.0 61000.0 No
4 False True False 40.0 63778.0 Yes

Output - Experiment 15 (continued)

5 True False False 35.0 58000.0 Yes
6 False False True 38.0 52000.0 No
7 True False False 48.0 79000.0 Yes
8 True False False 50.0 83000.0 No
9 True False False 370 670000 Yes
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
France Germany Spain Age Salary Purchased
0 True False False 44.0 72000.0 0
1 False False True 27.0 48000.0 1
2 False True False 30.0 54000.0 0
3 False False True 38.0 61000.0 0
4 False True False 40.0 63778.0 1
5 True False False 35.0 58000.0 1
6 False False True 38.0 52000.0 0
7 True False False 48.0 79000.0 1
8 True False False 50.0 83000.0 0
9 True False False 370 670000 1

Result - Experiment 15

Data preprocessing was carried out on the given data.